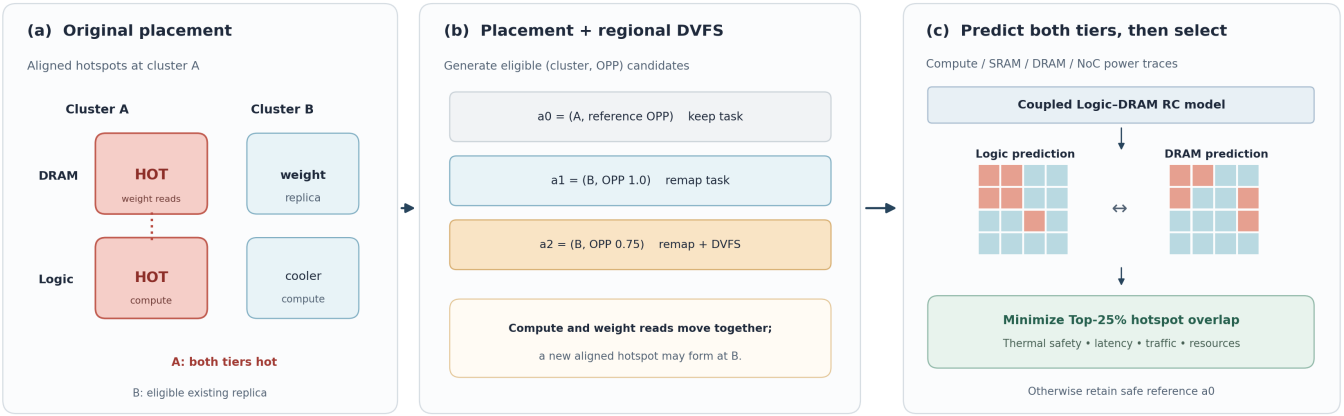

(a) Original placement
Aligned hotspots at cluster A
Cluster A
Cluster B
DRAM
HOT
weight reads
weight
replica
Logic
HOT
compute
cooler
compute
A: both tiers hot
B: eligible existing replica
(b) Placement + regional DVFS
Generate eligible (cluster, OPP) candidates
a0 = (A, reference OPP) keep task
a1 = (B, OPP 1.0) remap task
a2 = (B, OPP 0.75) remap + DVFS
Compute and weight reads move together;
a new aligned hotspot may form at B.
(c) Predict both tiers, then select
Compute / SRAM / DRAM / NoC power traces
Coupled Logic-DRAM RC model
Logic prediction
DRAM prediction
Minimize Top-25% hotspot overlap
Thermal safety • latency • traffic • resources
Otherwise retain safe reference a0